

Date	30 September 2026
Team ID	SWTID-2026-7220
Project Name	import data using transform maps(spreadsheet)
Maximum Marks	4 Marks

Project Design Phase-II

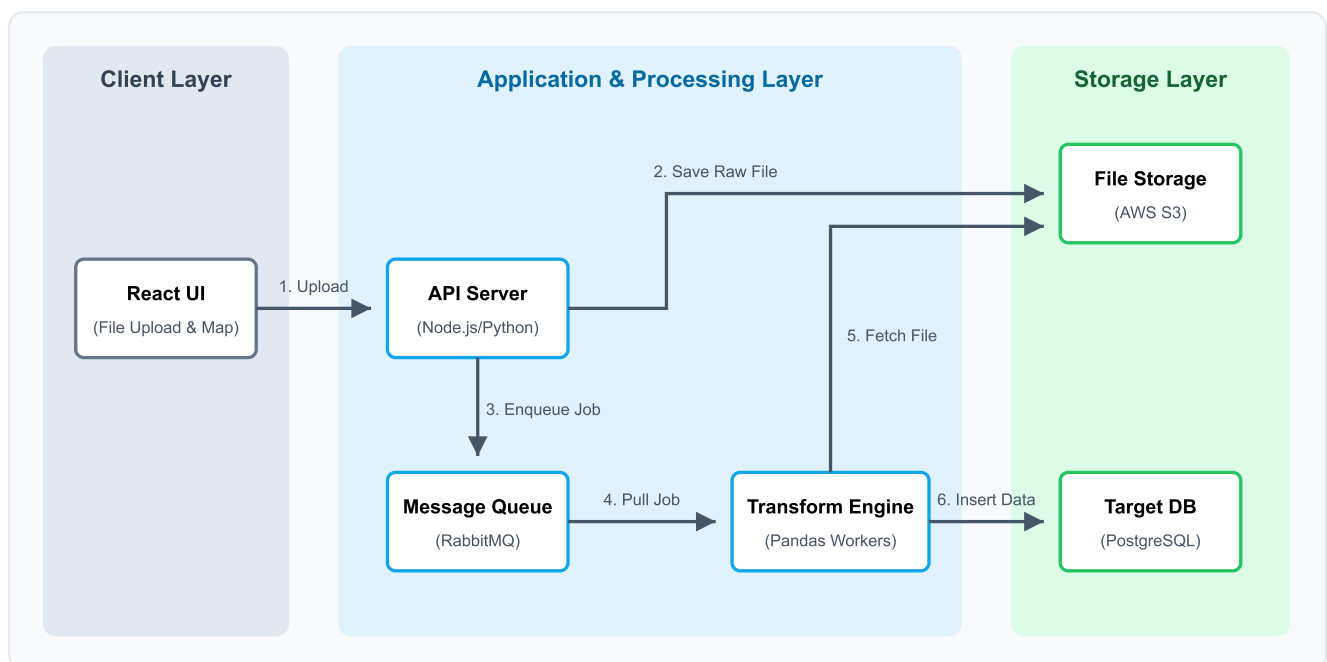
Technology Stack (Architecture & Stack)

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
1.	User Interface	Web interface for users to upload spreadsheets and define transform maps.	HTML5, CSS3, React.js
2.	Application Logic-1 (File Parsing)	Logic to extract data from uploaded .csv, .xls, .xlsx files.	Python (Pandas) / Node.js (SheetJS)
3.	Application Logic-2 (Transform Maps)	Engine to map spreadsheet columns to target database fields based on user rules.	Java / Python / Node.js
4.	Application Logic-3 (Batch Import)	Asynchronous processing of large datasets to prevent UI blocking during import.	RabbitMQ / Celery / Kafka
5.	Database	Target database where the transformed spreadsheet data is imported.	PostgreSQL / MySQL
6.	Cloud Database	Managed relational database service for high availability and backups.	Amazon RDS / Google Cloud SQL
7.	File Storage	Temporary storage for uploaded spreadsheets before processing.	AWS S3 / Local Filesystem
8.	External API-1	Integration to fetch spreadsheets directly from cloud storage.	Google Sheets API / Microsoft Graph API
9.	External API-2	Third-party API for data validation and cleansing during import.	Data Validation API (e.g., Loqate)
10.	Machine Learning Model	Auto-mapping feature to intelligently suggest column mappings based on headers.	Scikit-learn (NLP for column name matching)
11.	Infrastructure (Server / Cloud)	Application Deployment on Cloud Cloud Server Configuration:	Docker, Kubernetes (AWS EKS or GCP GKE)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for building the frontend, backend, and data processing.	React.js, Express.js, Python Pandas
2.	Security Implementations	Secure file uploads, data encryption at rest, and secure user access.	HTTPS/TLS, JWT Authentication, AES-256
3.	Scalable Architecture	Microservices architecture to handle multiple simultaneous large file uploads.	Docker Containers, Kubernetes
4.	Availability	Ensuring the import system is always available for end-users.	AWS Application Load Balancer, Multi-AZ
5.	Performance	Fast processing of large spreadsheets (100k+ rows) without memory leaks.	Redis (caching), Background workers

References:

- <https://c4model.com/>
- <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
- <https://www.ibm.com/cloud/architecture>
- <https://aws.amazon.com/architecture>
- <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>